

Income vs. Inflation: Understanding Housing Affordability in Canada Across Generations

Dennis Tat
University of Toronto
dennis.tat@mail.utoronto.ca

1 Introduction

1.1 Background

With the next federal election just around the corner, issues of affordability and housing access have taken center stage in public discourse. Canada has experienced a notable increase in the cost of living over the past decade, evident by the rising inflation. These economic pressures have outpaced the growth of salaries, making it increasingly difficult for many Canadian young adults entering the workforce to achieve financial stability. For this demographic, home ownership has become an increasingly distant prospect. At the same time, those who already own property have seen their real estate assets appreciate significantly in value, contributing to a growing sense of inequality between generations.

This widening gap has fueled a prevailing narrative that older generations had a far easier path to financial security and home ownership than today's youth. While anecdotal evidence and media commentary have popularized this belief, the goal of this study is to perform empirical analysis to determine the extent to which this perception is grounded in economic reality.

This study seeks to explore these dynamics by examining the relationship between inflation, housing prices, and salaries, in order to provide a clearer picture of how economic conditions have evolved over time.

1.2 Research Questions

To investigate the aforementioned imbalance in spending power between younger and older generations, this study will evaluate a multitude of economic factors. Several key questions will guide the analysis:

Has the median income of Canadians kept pace with inflation over time?

How has median income evolved across age groups, especially for younger versus older Canadians?

How does the change in income compare to the rise in housing prices?

Finally, can housing prices be predicted by analyzing the price trends of other consumer goods?

1.3 Data sets

This study draws from three datasets:

‘Income of individuals by age group, sex and income source, Canada, provinces and selected census metropolitan areas’ is a dataset that provides income data segmented by age group, gender, and province across Canada. It spans from 1972 to the present and serves as the primary source for analyzing trends in income over time, as well as comparing income growth across generations and between demographic groups.

‘Consumer Price Index, annual average, not seasonally adjusted’ features the Consumer Price index of many goods and services, ranging from 1914 to 2024. The CPI is a useful metric for measuring inflation, as it serves as a direct way track inflation and assess changes in purchasing power over time.

‘Analytical house prices indicators’ is similar to the above CPI data set, but it focuses specifically on average home prices throughout the years. Unlike the housing component of the CPI dataset, which factored in rental costs, this dataset reflects the actual purchase price of residential properties, making it more relevant for analyzing real estate affordability.

1.4 A Quick Note on Interpreting Price Indexes

The Consumer Price Index (CPI) measures the average change in the price of a specific set of goods and services over time. This value is always relative to a designated base year, which is assigned a CPI value of 100. Values for subsequent years indicate how prices have changed in comparison to that base year. For example, in the dataset used for this study, 1972 is the base year. If the CPI for eggs in 2024 is 250, it means that eggs are 2.5 times more expensive in 2024 than they were in 1972. The CPI thus provides a method for analyzing inflation and shifts in purchasing power across different time periods.

An index value was also added to the income and housing data set, and they can be interpreted in the same manner.

2 Methods

2.1 Data Wrangling and Cleaning

In this study, the primary metric used to measure inflation is the index value. Since this will be the value compared to across all data sets, we need to add it the income data set. First, a common base year must be selected, with the index value for that year standardized to 100. For the purposes of this analysis, the year 2002 has been chosen as the base year, as it approximates the average birth year of Generation Z, who are the demographic that this study is focusing on.

To convert our median salaries to price indexes for each demographic and year, we use the following formula

$$\text{Income Index} = \frac{[\text{Year}] \text{ Median Salary of Group X}}{2002 \text{ Median Salary of Group X}} \times 100$$

To ensure our price and housing index values are also standardized, the following formula is used

$$\text{Updated Price Index} = \frac{\text{Original Price Index}}{2002 \text{ Price Index}} \times 100$$

Which will not only ensure that the year 2002 will have values of 100, but also that every other year's price index will be relative to it.

Finally, all unnecessary variables were removed from the data sets, and the remaining variables which will be used from the three different data sets are as follows:

Variables	Type
Year	Numeric
Age Group	Factor
Sex	Factor
Median Income	Numeric
Income Index	Numeric

Table 1: Variables of the Income data set.

Variables	Type
Year	Numeric
Price Index	Numeric
Product	String

Table 2: Variables of the Price Index data set.

Variables	Type
Year	Numeric
Housing Index	Numeric

Table 3: Variables of the Housing Index data set.

Finally, in order to incorporate the CPI values of products in our models, items under the product variable were transformed into columns. This made it easier to quickly evaluate the CPI values of key items, such as mortgage interest costs and health care, on a per year basis.

Since each of the research questions require different information, data sets were merged and processed separately for each question.

2.2 Data Exploration Methods

The following tools were used for Data exploration:

ggplot2 in order visualize the relationship between different variables, as well as answering some of the basic research questions.

kable and kableExtra to summarize results such as largest and smallest values among the data sets.

plotly for interactive plots as seen on the website.

2.3 Modeling Methods

In order to predict housing price indexes, multiple models were used to accomplish this task.

2.3.1 Basic Linear Regression

To construct the linear regression model, products from the CPI data set were selected, based on how significant they were in the economy. These variables were food, mortgage interest, utility costs, gasoline, healthcare, dental, education, property tax, and transportation.

2.3.2 Generalized Additive Model

Using the same variables as described above, a GAM model was fit with a smooth function on the Year variable. During the exploratory data analysis, it was noted that the change in housing index was not linear with respect to the year. As such, a GAM model was used, as it is able to better represent this non-linear relationship.

2.3.3 Bagging

Bagging is a technique used to reduce variance in a predictive model. We use bootstrap samples of our training data set to train multiple decision trees. Once the trees have been trained, we can make predictions by taking the average of the decision tree predictions.

2.3.4 Random Forest

This technique is similar to bagging, as we also use bootstrap samples of our training data set to train decision trees. However, at each split of the tree, a random subset of predictors are selected, and the optimal predictor of the subset is used. This allows for the trees to be independent and decorrelated, as we are introducing some noise to ensure the trees are not going to be too similar.

3 Results

3.1 Exploratory Data Analysis

Before diving into our graphs and models, it's important to have a good understanding of the data that is being used.

Let's have a look at which demographics and years had the largest and smallest income indexes

Year	Age Group	Sex	Income Index
2020	55 to 64 years	Females	190.74
2022	55 to 64 years	Females	190.28
1977	15 to 24	Males	189.23
1979	15 to 24	Males	189.23
2021	55 to 64 years	Females	188.89
...	...	...	...
1976	65 years and over	Males	60.11
1976	65 years and over	Females	57.81
1977	65 years and over	Both sexes	57.45
1976	65 years and over	Both sexes	57.09
1977	65 years and over	Males	55.56

Table 4: Largest and Smallest Income Indexes.

An extremely interesting point to note is the years and demographics which had the largest income index are young men aged 15-24 in the 1970s, and older women aged 55-64 in the 2010s and 2020s. This is most because during the 1970s, it was more common for teenage boys to work, as well as the fact that

more women have joined the workforce compared to before. On the other hand, the lowest income index is mainly comprised of older people aged 55-64 in the 1970s. This is most likely because retirement benefits have improved by a lot since then. Its also important to note that this does not mean these people made the most or least amount of money, rather, the group that had the highest or lowest income within their own group.

Year	Product	Price Index
1995	Digital computing equipment and devices	390.0
2022	Fuel oil and other fuels	388.7
1996	Digital computing equipment and devices	357.7
2023	Fuel oil and other fuels	345.2
2024	Homeowners' home and mortgage insurance	342.9
2024	Cigarettes	323.37
...	...	...
1977	Cigarettes	13.0
1977	Tobacco products and smokers' supplies	12.9
1976	Air transportation	12.9
2024	Digital computing equipment and devices	12.7
1976	Cigarettes	12.0

Table 5: Largest and Smallest Consumer Price Indexes.

Moving on to the CPI data set, it appears that not only are cigarettes and tobacco more expensive now than ever, but that it also had its cheapest recorded prices in the 1970s. The price of nicotine products has been steadily increasing in order to encourage users to quit, which would explain the tripling in prices since 2002.

Additionally, oil in recent years is also at an all time high, most likely due to the ongoing conflicts around the world, as well as the Carbon Tax.

Conversely, we see that digital computing equipment has decreased by a multiple of 30 between 1990 and 2024, as computers and phones have become more accessible in recent years.

Year	Product	Price Index
1976	All items	31.1
2024	All items	160.9

Table 6: General Pricing Index Comparison

Finally, looking at the general price index which encapsulates all items, it can be deduced that a dollar in 1976 would be worth \$5.17 in 2024.

It should be no surprise that housing prices have also increased by a lot since the 70s, with prices peaking around recent years. Although these results appear to be aligned with the other products earlier, the housing data takes an average

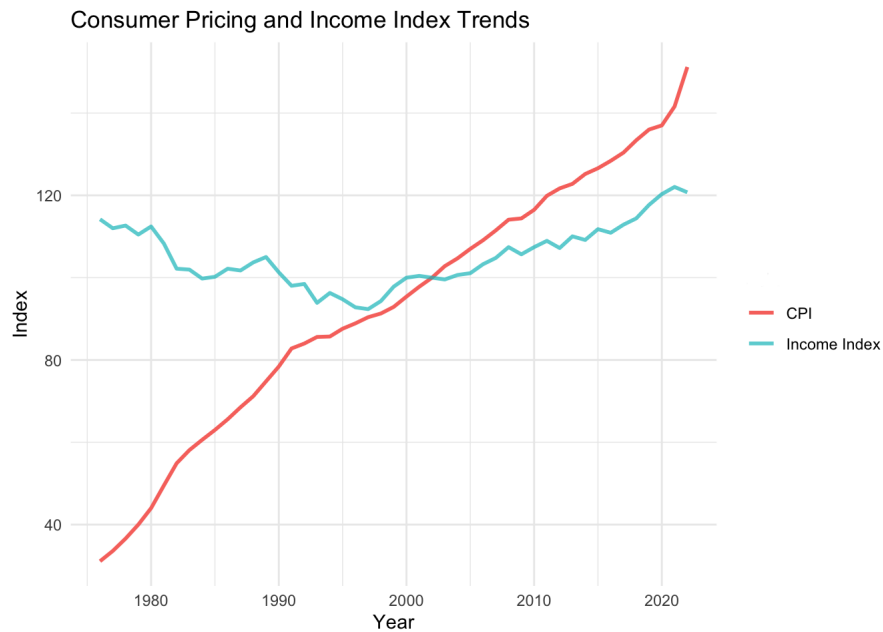
Year	Housing Index
2022	281.72
2024	267.56
2021	267.21
2023	265.95
2020	239.7
...	...
1985	66.33
1984	65.53
1972	60.48
1971	58.55
1970	57.38

Table 7: Largest and Smallest Housing Price Indexes.

across all of Canada. Observing data sets of Vancouver or Toronto would most likely yield more extreme results.

3.2 Graphing

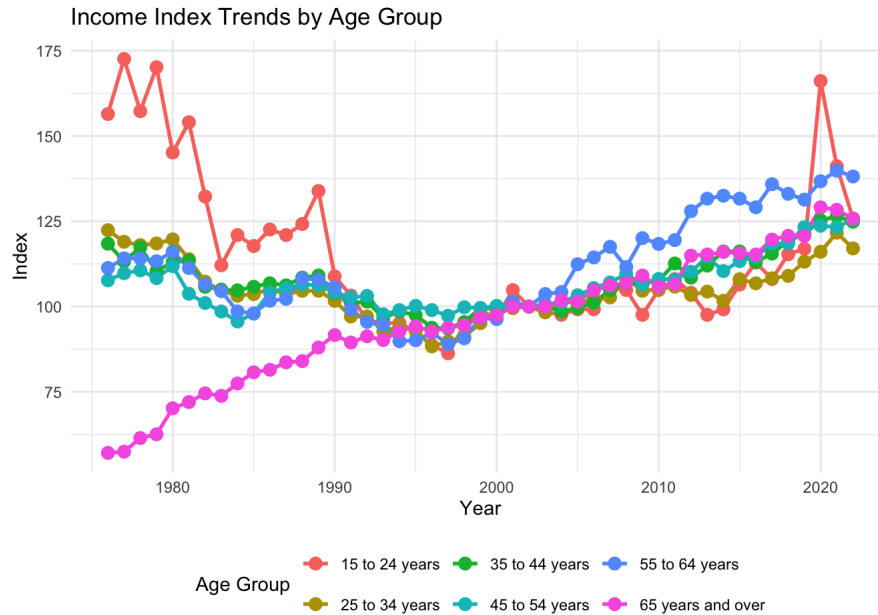
To determine whether the median annual income of Canadians have kept up with inflation, we can graph both the CPIs and Income indexes together.



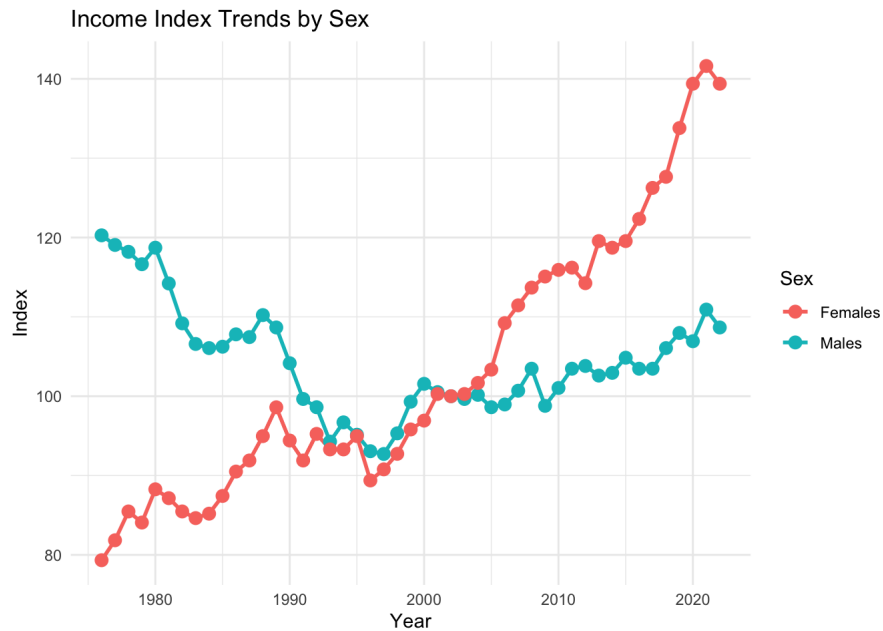
Note that while the CPI has been increasing steadily over the years, the annual

income has remained mostly stagnant. As such, young people today have much less purchasing power compared to others who began working in the 80s and 90s. Its also worth noting that the 2020 Covid-19 pandemic also led sharp rise to cost of living, which is evidenced from the steep slope of the graph.

Now lets switch our focus to the historical median income of Canadians across different demographics.

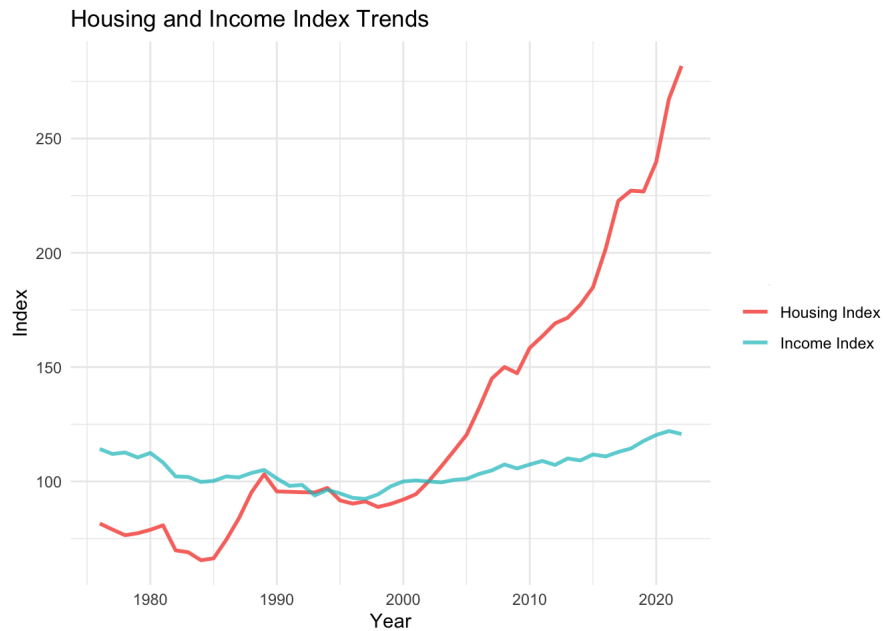


We see that older Canadians have had an income increase, whereas younger Canadians have an income decrease. Factors include less children working, seniors having later retirements, as well as retirement benefits in the 21st century. The spike in annual income for the youngest age group in 2020 was most likely due to the CESB grant that was given to students during the pandemic.



We also see that women have had an increase in their annual income, most likely due to the fact historically, less women were part of the workforce.

Lastly we look at housing. This is one of the products that has been hit the hardest in terms of inflation. Canada has been faced with a housing crisis for some time now, and it is especially prevalent in larger cities like Toronto and Vancouver, which are experiencing a large spike of population growth and a housing infrastructure that is unable to keep up.



Comparing the housing index with the income index of Canadians aged 25 - 54, it can be noted that these two values are moving further and further apart from each other. Had someone purchased a house in during the 90's, it would have nearly tripled in value since then. For Canadians who are looking to purchase a house today, under the same conditions of their parents or grand parents, they would likely need to earn over three times the median annual income, after accounting for our bracketed tax system as well as mortgage rates.

3.3 Modeling

3.3.1 Linear Regression

To model the housing price index changes, let's first start with the basic linear regression model.

Variable	Coefficient	Standard Error	P-Value
(Intercept)	614.4861	5379.6564	0.909618
Food	1.5400	0.5804	0.011287
Mortgage Interest Cost	-0.2958	0.3254	0.368779
Health Care	-2.4445	0.8199	0.004812
Property Taxes	1.9477	0.5138	0.000485
Year	-0.2863	2.7342	0.917107
Gasoline	0.2758	0.1476	0.068813
Transportation	-0.4112	0.6161	0.508181

Table 8: Summary of the linear regression model

Based on the results of this model summary, the most significant variables are property tax and health care.

Note that for each variable, holding all other variables constant, a one-unit change in the variable is associated with a change in the housing price index equal to the coefficient value.

3.3.2 Generalized Additive Model

Next, a GAM model was used, with smoothing on the Year variable.

Variable	Coefficient	Standard Error	P-Value
(Intercept)	276.8613	91.0782	0.004544
Food	-0.7203	0.5173	0.172897
Mortgage Interest Cost	-0.5207	0.1410	0.000777
Health Care	2.0720	0.9799	0.041915
Property Taxes	-1.6141	0.6598	0.019776
Gasoline	0.2139	0.1375	0.129108
Transportation	-0.7574	0.6128	0.224998
s(Year)	8.163	8.816	<2e-16

Table 9: Summary of the GAM model

Because the increases in the housing price index are not linear with respect to the year, using a GAM model may better capture this non-linear relationship.

3.3.3 Bagging and Random Forest

Using bagging and random forest resulted in rather weak models with high error values. None of the variables proved to be particularly valuable, as they all hovered between the 6-9% range. Moreover, the plots were difficult to interpret due to the fact that these values were too close to each other.

3.3.4 Model Evaluation

Finally, we observe the performance of the models.

Model	Error/MSE
Linear Regression	116.41576
GAM	12.64391
Bagging	66.61516
Random Forest	66.35038

Table 10: Error/MSE values of our 4 models

As expected, the GAM model had the lowest amount of error, due to the presence of non-linearity in the data. Another consequence of this non-linearity is the poor performance of the linear regression model.

We also see that the bagging and random forest models also under performed, due to the fact that the variables had low importance and purity.

4 Conclusions and Summary

4.1 Summary

Our analysis revealed that the median annual income of Canadians has struggled to keep pace with inflation over time. While median income has remained relatively stable within each demographic group, prices have not stopped rising. This is especially prevalent in the housing market, in which prices have increased at a rate more than 2.5 times higher than the growth in annual income since 2002. Based on our findings, we conclude that homeownership was more attainable in the 1980s and 1990s compared to more recent years.

We were also able to construct models to try and predict housing prices, using the price increases of other products. Of the many products analyzed, some of the most effective predictors were food, mortgage interest costs, property taxes, and health care. Overall, the most effective model to predict housing prices was the GAM model, as it was able to capture the non-linear relationship of housing prices throughout the years, which resulted in it having the lowest MSE.

4.2 Limitations

One key limitation of this analysis is the narrow scope of the data set, which only focused on pricing indexes. While this provides valuable information, it fails to capture the full complexity of the factors influencing economic conditions. Economic outcomes are impacted by a variety of factors, including employment rates, population growth, and global events. As such, the reliance on pricing

indexes alone oversimplifies the change in housing prices. Additionally, the limited size of the data meant that other modeling techniques such as XGBoost were unable to be used.

Another significant limitation arises from the geographical scope of the study. Because Canada is such a large country, there exist variations in pricing trends and economic conditions depending on the province. National-level pricing indexes may overly generalize the situation, particularly in urban centers and rural areas.